

## Solution of MGMT 670 Final Exam

Fall 2024

Name (Print): \_\_\_\_\_

Student ID No.: \_\_\_\_\_

Section: \_\_\_\_\_ 1: MWF 2:50 – 4:20 pm  
\_\_\_\_\_ 2: MWF 4:30 – 6:00 pm

EXAM Rules:

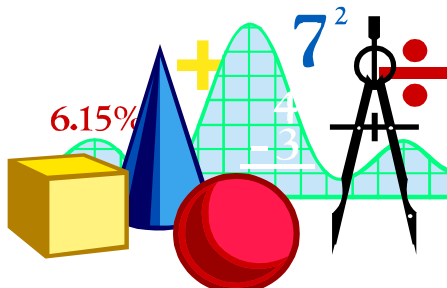
This is an open-book, open-notes exam. Bring your own calculator.  
**No cellphone. No internet. No cheating!**

### PART I. MULTIPLE CHOICES

1. Circle your answers on the exam.

### PART II. WORK PROBLEMS

1. Give neat, clean, and well organized answers.
2. Show your work to support your answers.



## Part I: Multiple Choice (15 Questions, 60 Points in Total)

Identify the letter of the choice that **best** completes the statement or answers the question.

Use the following information to answer the next two questions

A small manufacturing plant has sales offices located in four cities: Dallas, Seattle, Boston, and, Los Angeles. An analysis of the plants' account receivables has been used to develop the following contingency table for an arbitrary overdue invoice. For example, the probability that an overdue invoice is from the Dallas sales office and is less than 30 days overdue is 11%.

| Days overdue  | Dallas | Seattle | Boston | Los Angeles |
|---------------|--------|---------|--------|-------------|
| Under 30 days | 0.11   | 0.09    | 0.15   | 0.21        |
| 30-60 days    | 0.06   | 0.03    | 0.05   | 0.08        |
| 61-90 days    | 0.02   | 0.02    | 0.04   | 0.03        |
| Over 90 days  | 0.01   | 0.02    | 0.03   | 0.05        |

- What is the probability that an invoice is either from Los Angeles or is 61-90 days overdue?
  - 0.03
  - 0.37
  - 0.45
  - 0.48
- Given that an overdue invoice is from Boston, what's probability that it is under 30 days overdue?
  - 0.15
  - 0.27
  - 0.56
  - 0.68
- In regression analysis, which of the following is true:
  - Larger of  $R^2$  imply that the observations are more closely grouped about the average value of the dependent variable.
  - Multicollinearity means explanatory variables are highly correlated with the response variable.
  - When an additional independent variable is added to the model, the adjusted coefficient of multiple determination  $R_a^2$  will always be non-decreasing.
  - None of the above is correct.
- If a categorical variable has 4 levels (for example, season of a year = {Spring, Summer, Fall, Winter}), how many dummy variables are needed to fit the regression model?
  - 2
  - 3
  - 4
  - 5

5. Suppose the actual mean of a population is 30. A sample of size 40 is taken and the sample mean is computed as 42. If  $H_0: \mu \geq 25$ , and you decided not to reject the null hypothesis, it means
  - a. you have committed a Type I error.
  - b. you have committed a Type II error.
  - c. you have committed either Type I or Type II error.
  - d. you have committed neither Type I nor Type II error.
6. In testing  $H_0: \mu_1 - \mu_2 = 0$  against  $H_a: \mu_1 - \mu_2 \neq 0$ , we obtained the z-test statistic to be -1.87. Given that  $\text{Prob}(z < -1.87) = 0.03$ . Then which of the following conclusion is correct?
  - a. Reject  $H_0$  at  $\alpha = 5\%$  since p-value =  $0.03 < 0.05$
  - b. Cannot reject  $H_0$  at  $\alpha = 5\%$  since p-value =  $0.03 < 0.05$
  - c. Cannot reject  $H_0$  at  $\alpha = 1\%$  since p-value =  $0.03 > 0.01$
  - d. None of the above is correct.
7. The approximate mean and standard deviation of a company's invoice errors is \$42 and \$30, respectively. An auditor will come at the beginning of next month to examine the company's invoice errors. What is the probability that the average error per invoice in a sample of 36 invoices randomly selected by the auditor is less than or equal to \$50?
  - a. 0.9452
  - b. 0.5793
  - c. 0.4207
  - d. 0.0548
8. Suppose it is estimated that the proportion of people opposing the constitution on a certain aspect is between 30% and 45%. What is the minimum sample size needed to ensure that the margin of error is less than 2% at 98% confidence level?
  - a. 838
  - b. 1675
  - c. 2842
  - d. 3360
9. It is commonly believed that work stress has a negative impact on the personal lives of 75% of the restaurant employees. A sample of 100 employees of a restaurant finds that 68 answered "yes" when asked "Does work stress have a negative impact on your personal life?" We wish to see if the sample provides evidence that the percentage of restaurant employees affected by work stress is not 75%. At 5% level of significance, which of the following is correct?
  - a. Not enough data has been provided to conduct the hypothesis test
  - b. The null hypothesis should be rejected since the test statistic is -1.5 which is larger than the critical value -1.96
  - c. The null hypothesis should not be rejected since the test statistic is -1.5, and the test statistic lies between the critical values -1.96 and 1.96.
  - d. The null hypothesis should not be rejected since the test statistic is -1.62, and the test statistic lies between the critical values -1.96 and 1.96

Use the following information to answer the following two questions:

Michael, Nancy and Associates produce color printers. The demand for printers could be light, medium, or high with the following probabilities.

|                    | <b>Light Demand</b> | <b>Medium Demand</b> | <b>High Demand</b> |
|--------------------|---------------------|----------------------|--------------------|
| <b>Probability</b> | 0.4                 | 0.3                  | 0.3                |

The company has four production alternatives for the coming period. The payoffs (in millions of dollars) associated with the three alternatives are shown below:

|                      | <b>Light Demand</b> | <b>Medium Demand</b> | <b>High Demand</b> |
|----------------------|---------------------|----------------------|--------------------|
| <b>Alternative 1</b> | 18                  | 28                   | 20                 |
| <b>Alternative 2</b> | 25                  | 17                   | -5                 |
| <b>Alternative 3</b> | 3                   | 40                   | 16                 |
| <b>Alternative 4</b> | 5                   | 40                   | 20                 |

10. Based on expected values, which option will a rational decision maker choose?
  - a. Alternative 1
  - b. Alternative 2
  - c. Alternative 3
  - d. Alternative 4
11. What is the expected value of perfect information (in millions of dollars)?
  - a. 6.4
  - b. 21.6
  - c. 28
  - d. None of above

Use the following information to answer the following two questions:

USA Today reported (“Study finds better survival rates at ‘high volume’ hospitals”) that “high-volume” hospitals performed 77% of the bladder removal surgeries with the “low-volume” hospitals performing the rest. In the first two weeks after surgery, 3.1% of the patients at the “low-volume” hospital died compared to 0.7% at the high-volume hospitals.

12. What is the approximately the probability that a randomly chosen patient survives for more than two weeks after the bladder removal surgery?
  - a. 0.0123
  - b. 0.7646
  - c. 0.9875
  - d. 0.9946
13. Among 100 bladder removal patients, what is the approximately the probability that two will die?
  - a. 1.25%
  - b. 22.57%
  - c. 35.66%
  - d. 86.91%

*Use the following information to answer the following two questions:*

The IT director of a large company is concerned about the implication of computer idle time. According to the company data, the mean idle time of the computers in the company is 69 minutes. The system administrator examines the records of 41 computers randomly selected from the purchasing group and finds that their mean idle time is 71.2 minutes with a sample standard deviation of 8 minutes. The IT director wants to determine whether the mean idle time of the computers in the purchasing group is higher than the company average.

14. Which of the following statement is true?
- a. The test statistic = 1.761. There is enough evidence to conclude that the mean idle time for computers in the purchasing group is higher than the company average at  $\alpha = 1\%$  level.
  - b. The test statistic = 1.761. There is enough evidence to conclude that the mean idle time for computers in the purchasing group is higher than the company average at  $\alpha = 5\%$  level.
  - c. The test statistic = 0.275. There is not enough evidence to conclude that the mean idle time for computers in the purchasing group is higher than the company average at  $\alpha = 5\%$  level.
  - d. None of the above is true.
15. What should be the minimum average idle time of a sample of 41 computers for you to reject the null hypothesis at 5% level of significance (assuming that the sample standard deviation is 8 minutes)?
- a. 70.628
  - b. 71.104
  - c. 72.027
  - d. None of the above.

### Problem 1:

- a. Specify the null and alternate hypothesis. (2 pts)
- b. What is the test statistic? What distribution does it follow? verify any assumptions involved. (5 pts)
- c. What is the p-value of the test? Based on this, what is the conclusion of the test at  $\alpha=0.05$ ? What would be the suggestion for Wal-Mart's CEO? (3 pts)

### Problem 2:

A group of 35 randomly selected individuals participated in a study comparing a name-brand cartridge and a generic cartridge for printing machines. Each participant used both types of cartridges, and the number of pages printed with each was recorded. This approach was taken because variations in individual usage were expected, with some participants printing heavy graphics requiring more toner, while others primarily printed text documents. It is assumed that the number of pages printed with name-brand cartridges, the number of pages printed with generic cartridges, and the difference in pages printed between the two types follow a normal distribution across participants. The following data was collected:

| Person | Name-Brand | Generic |
|--------|------------|---------|
| 1      | 326        | 300     |
| 2      | 266        | 260     |
| 3      | 402        | 357     |
| ⋮      | ⋮          | ⋮       |
| 33     | 375        | 378     |
| 34     | 267        | 318     |
| 35     | 319        | 270     |

Below is the summary statistics based on this dataset:

|                                   | N  | Mean   | StDev | SE Mean |
|-----------------------------------|----|--------|-------|---------|
| Name-Brand                        | 35 | 342.77 | 54.65 | 9.24    |
| Generic                           | 35 | 323.43 | 57.25 | 9.68    |
| Difference (Name-Brand - Generic) | 35 | 19.34  | 63.9  | 10.8    |

We aim to test whether, on average, the name-brand cartridge produces more copies than the generic cartridge. Formulate the hypothesis (2 pts), compute the test statistic (3 pts) and critical value (2 pts), and provide the conclusion (2 pt). Use  $\alpha=0.05$ .

### Problem 3:

Sakura motors is designing a new car that will reduce greenhouse gas emissions for sale in markets where pollution is a major concern. Therefore, they conduct a study to determine the car characteristics that drive greenhouse gas emissions. The U.S. government publishes data on car models sold in the U.S. (fuelconomy.gov) which includes manufacturer, model, engine size (cylinders) and gas mileage (MPG or miles per gallon) for each car category. The data also includes the emissions of tons of greenhouse gases per year. Consumerreports.org provides data on acceleration in seconds to go from 0 to 60 miles per hour, which reflects model sluggishness, and measures of size, passengers and curb weight. The data is adapted from (BSCA, Cynthia Fraser).

| S. No. | Manfac.   | Model         | emit | MPG  | sec  | liters | wt    | hp  | cyl | passengers |
|--------|-----------|---------------|------|------|------|--------|-------|-----|-----|------------|
| 1      | acura     | mdx           | 9.7  | 20   | 8.2  | 3.5    | 2.066 | 265 | 6   | 7          |
| 2      | acura     | tl            | 7.9  | 24.5 | 6.7  | 3.2    | 1.617 | 270 | 6   | 5          |
| 3      | bmw       | 5series       | 7.7  | 24   | 7.4  | 3      | 1.656 | 215 | 6   | 5          |
| 4      | chevrolet | trailblazer   | 11.8 | 15.5 | 8.3  | 6      | 2.114 | 291 | 8   | 5          |
| 5      | dodge     | durango       | 11.4 | 15.5 | 7.6  | 5.7    | 2.42  | 210 | 8   | 7          |
| ⋮      | ⋮         | ⋮             | ⋮    | ⋮    | ⋮    | ⋮      | ⋮     | ⋮   | ⋮   | ⋮          |
| 29     | toyota    | Sequoia       | 11.5 | 16.5 | 9.7  | 4.7    | 2.395 | 273 | 8   | 8          |
| 30     | toyota    | Sienna        | 8.5  | 22.5 | 8.6  | 3.3    | 1.98  | 230 | 6   | 8          |
| 31     | toyota    | Tundra        | 11.4 | 16.5 | 8.8  | 4      | 2.311 | 245 | 6   | 6          |
| 32     | vw        | Toureg        | 11.9 | 18.5 | 11.9 | 4.2    | 2.363 | 240 | 8   | 5          |
| 33     | honda     | accord hybrid | 6.6  | 33   | 6.9  | 2.4    | 1.576 | 255 | 4   | 5          |
| 34     | Lexus     | RX400 hybrid  | 6.4  | 29   | 8.8  | 3.5    | 1.905 | 268 | 6   | 4          |

Thirty four imported/domestic cars including subcompacts, compacts, intermediates, full-size sedans, wagons, SUVs and pickups were selected for this dataset. The characteristics of the cars studied include emissions in tons of CO<sub>2</sub> released (*emit*), miles per gallon (*MPG*), seconds to accelerate from 0 to 60 miles per hour (*sec*), engine size (*liters*), weight in thousands of kilograms (*wt*), horsepower (*hp*), cylinders (*cyl*), and passenger capacity (*passengers*). We aim to use a linear regression model to predict *emit* using *MPG*, *sec*, *liters*, *wt*, *hp*, *cyl*, and *passengers*.

The following MINITAB regression reports were created.

#### Correlations: emit, MPG, sec, liters, wt, hp, cyl, passengers

|            | emit   | MPG    | sec    | liters | wt    | hp    | cyl   |
|------------|--------|--------|--------|--------|-------|-------|-------|
| MPG        | -0.936 |        |        |        |       |       |       |
| sec        | 0.144  | -0.049 |        |        |       |       |       |
| liters     | 0.835  | -0.810 | -0.171 |        |       |       |       |
| wt         | 0.828  | -0.769 | -0.013 | 0.835  |       |       |       |
| hp         | 0.553  | -0.529 | -0.363 | 0.763  | 0.718 |       |       |
| cyl        | 0.764  | -0.744 | -0.189 | 0.924  | 0.808 | 0.771 |       |
| passengers | 0.540  | -0.526 | -0.049 | 0.593  | 0.703 | 0.545 | 0.602 |

Cell Contents: Pearson correlation



### Regression Analysis: emit versus MPG, sec, ...

The regression equation is

emit = 9.16 - 0.226 MPG + 0.229 sec + 1.20 wt - 0.085 passengers - 0.00052 hp - 0.035 cyl + 0.414 liters

| Predictor  | Coef      | SE Coef  | T     | P     |
|------------|-----------|----------|-------|-------|
| Constant   | 9.165     | 1.904    | 4.81  | 0.000 |
| MPG        | -0.22612  | 0.03656  | -6.18 | 0.000 |
| sec        | 0.22915   | 0.09865  | 2.32  | 0.028 |
| wt         | 1.1992    | 0.6499   | 1.85  | 0.076 |
| passengers | -0.0855   | 0.1198   | -0.71 | 0.482 |
| hp         | -0.000519 | 0.003688 | -0.14 | 0.889 |
| cyl        | -0.0350   | 0.1883   | -0.19 | 0.854 |
| liters     | 0.4142    | 0.2934   | 1.41  | ???   |

S = 0.643697    **R-Sq = ???**    R-Sq(adj) = 90.8%

Analysis of Variance

| Source         | DF | SS      | MS     | <b>F</b>   | <b>P</b>   |
|----------------|----|---------|--------|------------|------------|
| Regression     | 7  | 138.443 | 19.778 | <b>???</b> | <b>???</b> |
| Residual Error | 26 | 10.773  | 0.414  |            |            |
| Total          | 33 | 149.216 |        |            |            |

### Regression Analysis: liters versus MPG, sec, wt, hp, cyl, passengers

The regression equation is

liters = 1.51 - 0.0533 MPG - 0.0371 sec + 0.427 wt + 0.0022 hp + 0.412 cyl - 0.0283 passengers

| Predictor  | Coef     | SE Coef  | T     | P     |
|------------|----------|----------|-------|-------|
| Constant   | 1.515    | 1.214    | 1.25  | 0.223 |
| MPG        | -0.05333 | 0.02167  | -2.46 | 0.021 |
| sec        | -0.03710 | 0.06431  | -0.58 | 0.569 |
| wt         | 0.4273   | 0.4183   | 1.02  | 0.316 |
| hp         | 0.002225 | 0.002381 | 0.93  | 0.358 |
| cyl        | 0.41200  | 0.09469  | 4.35  | 0.000 |
| passengers | -0.02833 | 0.07836  | -0.36 | 0.721 |

S = 0.422195    R-Sq = 90.2%    R-Sq(adj) = 88.0%

### Regression Analysis: liters versus emit, MPG, ...

The regression equation is

liters = - 0.17 + 0.172 emit - 0.0107 MPG - 0.0738 sec + 0.191 wt + 0.00216 hp + 0.389 cyl - 0.0116 passengers

| Predictor  | Coef     | SE Coef  | T     | P     |
|------------|----------|----------|-------|-------|
| Constant   | -0.168   | 1.686    | -0.10 | 0.921 |
| emit       | 0.1719   | 0.1218   | 1.41  | 0.170 |
| MPG        | -0.01068 | 0.03696  | -0.29 | 0.775 |
| sec        | -0.07384 | 0.06831  | -1.08 | 0.290 |
| wt         | 0.1908   | 0.4436   | 0.43  | 0.671 |
| hp         | 0.002156 | 0.002339 | 0.92  | 0.365 |
| cyl        | 0.38869  | 0.09445  | 4.12  | 0.000 |
| passengers | -0.01163 | 0.07786  | -0.15 | 0.882 |

S = 0.414645    R-Sq = 90.9%    R-Sq(adj) = 88.4%

- a. Based on the first regression model, calculate the predicted emissions and the residual associated with the following car. (3 pts)

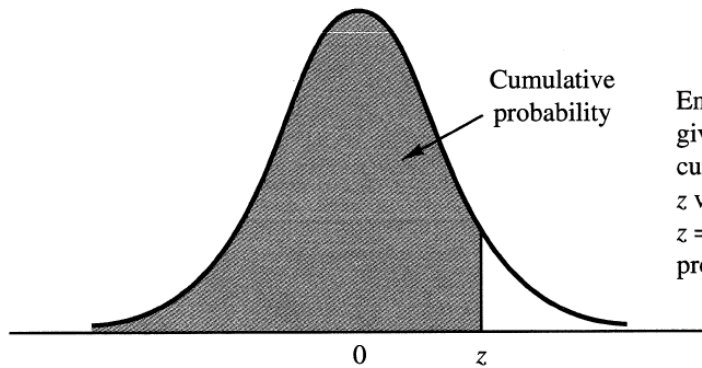
| S. No. | Manfac. | Model | emit | MPG | sec | liters | wt    | hp  | cyl | passengers |
|--------|---------|-------|------|-----|-----|--------|-------|-----|-----|------------|
| 1      | acura   | mdx   | 9.7  | 20  | 8.2 | 3.5    | 2.066 | 265 | 6   | 7          |

- b. Compute  $R-Sq$  of the first regression model. Interpret this  $R-Sq$  in a short sentence. (3 pts)

- c. Conduct the F test for the overall fit of the first regression model. State the hypotheses to be tested (2 pts), the test statistic (2 pts), the critical value (1 pt), and your conclusion (1 pt). Use  $\alpha=0.05$ .

- d. Based on the output from the first regression model, compute 95% confidence interval for the coefficient of “*liters*”. Can we reject the null hypothesis that this coefficient is 0 with  $\alpha=0.05$ ? (4 pts)
- e. Calculate VIF of the predictor “*liters*”. (3 pts)
- f. When we fit the model:  $\text{Emit} = \beta_0 + \beta_1 (\text{liters}) + \varepsilon$ , the F test is significant and indicates that “*liters*” is significant for predicting “*emit*”. Explain why this result might differ from results in previous problems. (2 pts)

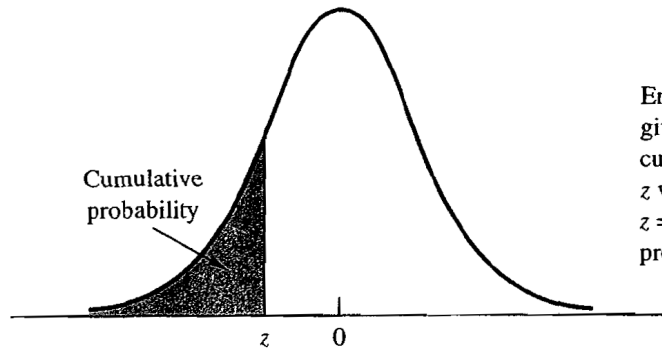
# CUMULATIVE PROBABILITIES FOR THE STANDARD NORMAL DISTRIBUTION



Entries in the table give the area under the curve to the left of the  $z$  value. For example, for  $z = 1.25$ , the cumulative probability is .8944.

| $z$ | .00   | .01   | .02   | .03   | .04   | .05   | .06   | .07   | .08   | .09   |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| .0  | .5000 | .5040 | .5080 | .5120 | .5160 | .5199 | .5239 | .5279 | .5319 | .5359 |
| .1  | .5398 | .5438 | .5478 | .5517 | .5557 | .5596 | .5636 | .5675 | .5714 | .5753 |
| .2  | .5793 | .5832 | .5871 | .5910 | .5948 | .5987 | .6026 | .6064 | .6103 | .6141 |
| .3  | .6179 | .6217 | .6255 | .6293 | .6331 | .6368 | .6406 | .6443 | .6480 | .6517 |
| .4  | .6554 | .6591 | .6628 | .6664 | .6700 | .6736 | .6772 | .6808 | .6844 | .6879 |
| .5  | .6915 | .6950 | .6985 | .7019 | .7054 | .7088 | .7123 | .7157 | .7190 | .7224 |
| .6  | .7257 | .7291 | .7324 | .7357 | .7389 | .7422 | .7454 | .7486 | .7517 | .7549 |
| .7  | .7580 | .7611 | .7642 | .7673 | .7704 | .7734 | .7764 | .7794 | .7823 | .7852 |
| .8  | .7881 | .7910 | .7939 | .7967 | .7995 | .8023 | .8051 | .8078 | .8106 | .8133 |
| .9  | .8159 | .8186 | .8212 | .8238 | .8264 | .8289 | .8315 | .8340 | .8365 | .8389 |
| 1.0 | .8413 | .8438 | .8461 | .8485 | .8508 | .8531 | .8554 | .8577 | .8599 | .8621 |
| 1.1 | .8643 | .8665 | .8686 | .8708 | .8729 | .8749 | .8770 | .8790 | .8810 | .8830 |
| 1.2 | .8849 | .8869 | .8888 | .8907 | .8925 | .8944 | .8962 | .8980 | .8997 | .9015 |
| 1.3 | .9032 | .9049 | .9066 | .9082 | .9099 | .9115 | .9131 | .9147 | .9162 | .9177 |
| 1.4 | .9192 | .9207 | .9222 | .9236 | .9251 | .9265 | .9279 | .9292 | .9306 | .9319 |
| 1.5 | .9332 | .9345 | .9357 | .9370 | .9382 | .9394 | .9406 | .9418 | .9429 | .9441 |
| 1.6 | .9452 | .9463 | .9474 | .9484 | .9495 | .9505 | .9515 | .9525 | .9535 | .9545 |
| 1.7 | .9554 | .9564 | .9573 | .9582 | .9591 | .9599 | .9608 | .9616 | .9625 | .9633 |
| 1.8 | .9641 | .9649 | .9656 | .9664 | .9671 | .9678 | .9686 | .9693 | .9699 | .9706 |
| 1.9 | .9713 | .9719 | .9726 | .9732 | .9738 | .9744 | .9750 | .9756 | .9761 | .9767 |
| 2.0 | .9772 | .9778 | .9783 | .9788 | .9793 | .9798 | .9803 | .9808 | .9812 | .9817 |
| 2.1 | .9821 | .9826 | .9830 | .9834 | .9838 | .9842 | .9846 | .9850 | .9854 | .9857 |
| 2.2 | .9861 | .9864 | .9868 | .9871 | .9875 | .9878 | .9881 | .9884 | .9887 | .9890 |
| 2.3 | .9893 | .9896 | .9898 | .9901 | .9904 | .9906 | .9909 | .9911 | .9913 | .9913 |
| 2.4 | .9918 | .9920 | .9922 | .9925 | .9927 | .9929 | .9931 | .9932 | .9934 | .9936 |
| 2.5 | .9938 | .9940 | .9941 | .9943 | .9945 | .9946 | .9948 | .9949 | .9951 | .9952 |
| 2.6 | .9953 | .9955 | .9956 | .9957 | .9959 | .9960 | .9961 | .9962 | .9963 | .9964 |
| 2.7 | .9965 | .9966 | .9967 | .9968 | .9969 | .9970 | .9971 | .9972 | .9973 | .9974 |
| 2.8 | .9974 | .9975 | .9976 | .9977 | .9977 | .9978 | .9979 | .9979 | .9980 | .9981 |
| 2.9 | .9981 | .9982 | .9982 | .9983 | .9984 | .9984 | .9985 | .9985 | .9986 | .9986 |
| 3.0 | .9986 | .9987 | .9987 | .9988 | .9988 | .9989 | .9989 | .9989 | .9990 | .9990 |

# CUMULATIVE PROBABILITIES FOR THE STANDARD NORMAL DISTRIBUTION



Entries in this table give the area under the curve to the left of the  $z$  value. For example, for  $z = -.85$ , the cumulative probability is .1977.

| $z$  | .00   | .01   | .02   | .03   | .04   | .05   | .06   | .07   | .08   | .09   |
|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| -3.0 | .0013 | .0013 | .0013 | .0012 | .0012 | .0011 | .0011 | .0011 | .0010 | .0010 |
| -2.9 | .0019 | .0018 | .0018 | .0017 | .0016 | .0016 | .0015 | .0015 | .0014 | .0014 |
| -2.8 | .0026 | .0025 | .0024 | .0023 | .0023 | .0022 | .0021 | .0021 | .0020 | .0019 |
| -2.7 | .0035 | .0034 | .0033 | .0032 | .0031 | .0030 | .0029 | .0028 | .0027 | .0026 |
| -2.6 | .0047 | .0045 | .0044 | .0043 | .0041 | .0040 | .0039 | .0038 | .0037 | .0036 |
| -2.5 | .0062 | .0060 | .0059 | .0057 | .0055 | .0054 | .0052 | .0051 | .0049 | .0048 |
| -2.4 | .0082 | .0080 | .0078 | .0075 | .0073 | .0071 | .0069 | .0068 | .0066 | .0064 |
| -2.3 | .0107 | .0104 | .0102 | .0099 | .0096 | .0094 | .0091 | .0089 | .0087 | .0084 |
| -2.2 | .0139 | .0136 | .0132 | .0129 | .0125 | .0122 | .0119 | .0116 | .0113 | .0110 |
| -2.1 | .0179 | .0174 | .0170 | .0166 | .0162 | .0158 | .0154 | .0150 | .0146 | .0143 |
| -2.0 | .0228 | .0222 | .0217 | .0212 | .0207 | .0202 | .0197 | .0192 | .0188 | .0183 |
| -1.9 | .0287 | .0281 | .0274 | .0268 | .0262 | .0256 | .0250 | .0244 | .0239 | .0233 |
| -1.8 | .0359 | .0351 | .0344 | .0336 | .0329 | .0322 | .0314 | .0307 | .0301 | .0294 |
| -1.7 | .0446 | .0436 | .0427 | .0418 | .0409 | .0401 | .0392 | .0384 | .0375 | .0367 |
| -1.6 | .0548 | .0537 | .0526 | .0516 | .0505 | .0495 | .0485 | .0475 | .0465 | .0455 |
| -1.5 | .0668 | .0655 | .0643 | .0630 | .0618 | .0606 | .0594 | .0582 | .0571 | .0559 |
| -1.4 | .0808 | .0793 | .0778 | .0764 | .0749 | .0735 | .0721 | .0708 | .0694 | .0681 |
| -1.3 | .0968 | .0951 | .0934 | .0918 | .0901 | .0885 | .0869 | .0853 | .0838 | .0823 |
| -1.2 | .1151 | .1131 | .1112 | .1093 | .1075 | .1056 | .1038 | .1020 | .1003 | .0985 |
| -1.1 | .1357 | .1335 | .1314 | .1292 | .1271 | .1251 | .1230 | .1210 | .1190 | .1170 |
| -1.0 | .1587 | .1562 | .1539 | .1515 | .1492 | .1469 | .1446 | .1423 | .1401 | .1379 |
| -.9  | .1841 | .1814 | .1788 | .1762 | .1736 | .1711 | .1685 | .1660 | .1635 | .1611 |
| -.8  | .2119 | .2090 | .2061 | .2033 | .2005 | .1977 | .1949 | .1922 | .1894 | .1867 |
| -.7  | .2420 | .2389 | .2358 | .2327 | .2296 | .2266 | .2236 | .2206 | .2177 | .2148 |
| -.6  | .2743 | .2709 | .2676 | .2643 | .2611 | .2578 | .2546 | .2514 | .2483 | .2451 |
| -.5  | .3085 | .3050 | .3015 | .2981 | .2946 | .2912 | .2877 | .2843 | .2810 | .2776 |
| -.4  | .3446 | .3409 | .3372 | .3336 | .3300 | .3264 | .3228 | .3192 | .3156 | .3121 |
| -.3  | .3821 | .3783 | .3745 | .3707 | .3669 | .3632 | .3594 | .3557 | .3520 | .3483 |
| -.2  | .4207 | .4168 | .4129 | .4090 | .4052 | .4013 | .3974 | .3936 | .3897 | .3859 |
| -.1  | .4602 | .4562 | .4522 | .4483 | .4443 | .4404 | .4364 | .4325 | .4286 | .4247 |
| -0   | .5000 | .4960 | .4920 | .4880 | .4840 | .4801 | .4761 | .4721 | .4681 | .4641 |

**t Table**

| cum. prob<br>one-tail<br>two-tails | $t_{.50}$               | $t_{.75}$ | $t_{.80}$ | $t_{.85}$ | $t_{.90}$ | $t_{.95}$ | $t_{.975}$ | $t_{.99}$ | $t_{.995}$ | $t_{.999}$ | $t_{.9995}$ |
|------------------------------------|-------------------------|-----------|-----------|-----------|-----------|-----------|------------|-----------|------------|------------|-------------|
|                                    | 0.50                    | 0.25      | 0.20      | 0.15      | 0.10      | 0.05      | 0.025      | 0.01      | 0.005      | 0.001      | 0.0005      |
|                                    | 1.00                    | 0.50      | 0.40      | 0.30      | 0.20      | 0.10      | 0.05       | 0.02      | 0.01       | 0.002      | 0.001       |
| df                                 |                         |           |           |           |           |           |            |           |            |            |             |
| 1                                  | 0.000                   | 1.000     | 1.376     | 1.963     | 3.078     | 6.314     | 12.71      | 31.82     | 63.66      | 318.31     | 636.62      |
| 2                                  | 0.000                   | 0.816     | 1.061     | 1.386     | 1.886     | 2.920     | 4.303      | 6.965     | 9.925      | 22.327     | 31.599      |
| 3                                  | 0.000                   | 0.765     | 0.978     | 1.250     | 1.638     | 2.353     | 3.182      | 4.541     | 5.841      | 10.215     | 12.924      |
| 4                                  | 0.000                   | 0.741     | 0.941     | 1.190     | 1.533     | 2.132     | 2.776      | 3.747     | 4.604      | 7.173      | 8.610       |
| 5                                  | 0.000                   | 0.727     | 0.920     | 1.156     | 1.476     | 2.015     | 2.571      | 3.365     | 4.032      | 5.893      | 6.869       |
| 6                                  | 0.000                   | 0.718     | 0.906     | 1.134     | 1.440     | 1.943     | 2.447      | 3.143     | 3.707      | 5.208      | 5.959       |
| 7                                  | 0.000                   | 0.711     | 0.896     | 1.119     | 1.415     | 1.895     | 2.365      | 2.998     | 3.499      | 4.785      | 5.408       |
| 8                                  | 0.000                   | 0.706     | 0.889     | 1.108     | 1.397     | 1.860     | 2.306      | 2.896     | 3.355      | 4.501      | 5.041       |
| 9                                  | 0.000                   | 0.703     | 0.883     | 1.100     | 1.383     | 1.833     | 2.262      | 2.821     | 3.250      | 4.297      | 4.781       |
| 10                                 | 0.000                   | 0.700     | 0.879     | 1.093     | 1.372     | 1.812     | 2.228      | 2.764     | 3.169      | 4.144      | 4.587       |
| 11                                 | 0.000                   | 0.697     | 0.876     | 1.088     | 1.363     | 1.796     | 2.201      | 2.718     | 3.106      | 4.025      | 4.437       |
| 12                                 | 0.000                   | 0.695     | 0.873     | 1.083     | 1.356     | 1.782     | 2.179      | 2.681     | 3.055      | 3.930      | 4.318       |
| 13                                 | 0.000                   | 0.694     | 0.870     | 1.079     | 1.350     | 1.771     | 2.160      | 2.650     | 3.012      | 3.852      | 4.221       |
| 14                                 | 0.000                   | 0.692     | 0.868     | 1.076     | 1.345     | 1.761     | 2.145      | 2.624     | 2.977      | 3.787      | 4.140       |
| 15                                 | 0.000                   | 0.691     | 0.866     | 1.074     | 1.341     | 1.753     | 2.131      | 2.602     | 2.947      | 3.733      | 4.073       |
| 16                                 | 0.000                   | 0.690     | 0.865     | 1.071     | 1.337     | 1.746     | 2.120      | 2.583     | 2.921      | 3.686      | 4.015       |
| 17                                 | 0.000                   | 0.689     | 0.863     | 1.069     | 1.333     | 1.740     | 2.110      | 2.567     | 2.898      | 3.646      | 3.965       |
| 18                                 | 0.000                   | 0.688     | 0.862     | 1.067     | 1.330     | 1.734     | 2.101      | 2.552     | 2.878      | 3.610      | 3.922       |
| 19                                 | 0.000                   | 0.688     | 0.861     | 1.066     | 1.328     | 1.729     | 2.093      | 2.539     | 2.861      | 3.579      | 3.883       |
| 20                                 | 0.000                   | 0.687     | 0.860     | 1.064     | 1.325     | 1.725     | 2.086      | 2.528     | 2.845      | 3.552      | 3.850       |
| 21                                 | 0.000                   | 0.686     | 0.859     | 1.063     | 1.323     | 1.721     | 2.080      | 2.518     | 2.831      | 3.527      | 3.819       |
| 22                                 | 0.000                   | 0.686     | 0.858     | 1.061     | 1.321     | 1.717     | 2.074      | 2.508     | 2.819      | 3.505      | 3.792       |
| 23                                 | 0.000                   | 0.685     | 0.858     | 1.060     | 1.319     | 1.714     | 2.069      | 2.500     | 2.807      | 3.485      | 3.768       |
| 24                                 | 0.000                   | 0.685     | 0.857     | 1.059     | 1.318     | 1.711     | 2.064      | 2.492     | 2.797      | 3.467      | 3.745       |
| 25                                 | 0.000                   | 0.684     | 0.856     | 1.058     | 1.316     | 1.708     | 2.060      | 2.485     | 2.787      | 3.450      | 3.725       |
| 26                                 | 0.000                   | 0.684     | 0.856     | 1.058     | 1.315     | 1.706     | 2.056      | 2.479     | 2.779      | 3.435      | 3.707       |
| 27                                 | 0.000                   | 0.684     | 0.855     | 1.057     | 1.314     | 1.703     | 2.052      | 2.473     | 2.771      | 3.421      | 3.690       |
| 28                                 | 0.000                   | 0.683     | 0.855     | 1.056     | 1.313     | 1.701     | 2.048      | 2.467     | 2.763      | 3.408      | 3.674       |
| 29                                 | 0.000                   | 0.683     | 0.854     | 1.055     | 1.311     | 1.699     | 2.045      | 2.462     | 2.756      | 3.396      | 3.659       |
| 30                                 | 0.000                   | 0.683     | 0.854     | 1.055     | 1.310     | 1.697     | 2.042      | 2.457     | 2.750      | 3.385      | 3.646       |
| 40                                 | 0.000                   | 0.681     | 0.851     | 1.050     | 1.303     | 1.684     | 2.021      | 2.423     | 2.704      | 3.307      | 3.551       |
| 60                                 | 0.000                   | 0.679     | 0.848     | 1.045     | 1.296     | 1.671     | 2.000      | 2.390     | 2.660      | 3.232      | 3.460       |
| 80                                 | 0.000                   | 0.678     | 0.846     | 1.043     | 1.292     | 1.664     | 1.990      | 2.374     | 2.639      | 3.195      | 3.416       |
| 100                                | 0.000                   | 0.677     | 0.845     | 1.042     | 1.290     | 1.660     | 1.984      | 2.364     | 2.626      | 3.174      | 3.390       |
| 1000                               | 0.000                   | 0.675     | 0.842     | 1.037     | 1.282     | 1.646     | 1.962      | 2.330     | 2.581      | 3.098      | 3.300       |
| <b>z</b>                           | 0.000                   | 0.674     | 0.842     | 1.036     | 1.282     | 1.645     | 1.960      | 2.326     | 2.576      | 3.090      | 3.291       |
|                                    | 0%                      | 50%       | 60%       | 70%       | 80%       | 90%       | 95%        | 98%       | 99%        | 99.8%      | 99.9%       |
|                                    | <b>Confidence Level</b> |           |           |           |           |           |            |           |            |            |             |

## F Table

| Table of $F_{.05}$ Values            |                              |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |          |
|--------------------------------------|------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|
| Denominator<br>Degrees<br>of Freedom | Numerator Degrees of Freedom |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |          |
|                                      | 1                            | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    | 12    | 15    | 20    | 24    | 30    | 40    | 60    | 120   | $\infty$ |
| 1                                    | 161.4                        | 199.5 | 215.7 | 224.6 | 230.2 | 234.0 | 236.8 | 238.9 | 240.5 | 241.9 | 243.9 | 245.9 | 248.0 | 249.1 | 250.1 | 251.1 | 252.2 | 253.3 | 254.3    |
| 2                                    | 18.51                        | 19.00 | 19.16 | 19.25 | 19.30 | 19.33 | 19.35 | 19.37 | 19.38 | 19.40 | 19.41 | 19.43 | 19.45 | 19.45 | 19.46 | 19.47 | 19.48 | 19.49 | 19.50    |
| 3                                    | 10.13                        | 9.55  | 9.28  | 9.12  | 9.01  | 8.94  | 8.89  | 8.85  | 8.81  | 8.79  | 8.74  | 8.70  | 8.66  | 8.64  | 8.62  | 8.59  | 8.57  | 8.55  | 8.53     |
| 4                                    | 7.71                         | 6.94  | 6.59  | 6.39  | 6.26  | 6.16  | 6.09  | 6.04  | 6.00  | 5.96  | 5.91  | 5.86  | 5.80  | 5.77  | 5.75  | 5.72  | 5.69  | 5.66  | 5.63     |
| 5                                    | 6.61                         | 5.79  | 5.41  | 5.19  | 5.05  | 4.95  | 4.88  | 4.82  | 4.77  | 4.74  | 4.68  | 4.62  | 4.56  | 4.53  | 4.50  | 4.46  | 4.43  | 4.40  | 4.36     |
| 6                                    | 5.99                         | 5.14  | 4.76  | 4.53  | 4.39  | 4.28  | 4.21  | 4.15  | 4.10  | 4.06  | 4.00  | 3.94  | 3.87  | 3.84  | 3.81  | 3.77  | 3.74  | 3.70  | 3.67     |
| 7                                    | 5.59                         | 4.74  | 4.35  | 4.12  | 3.97  | 3.87  | 3.79  | 3.73  | 3.68  | 3.64  | 3.57  | 3.51  | 3.44  | 3.41  | 3.38  | 3.34  | 3.30  | 3.27  | 3.23     |
| 8                                    | 5.32                         | 4.46  | 4.07  | 3.84  | 3.69  | 3.58  | 3.50  | 3.44  | 3.39  | 3.35  | 3.28  | 3.22  | 3.15  | 3.12  | 3.08  | 3.04  | 3.01  | 2.97  | 2.93     |
| 9                                    | 5.12                         | 4.26  | 3.86  | 3.63  | 3.48  | 3.37  | 3.29  | 3.23  | 3.18  | 3.14  | 3.07  | 3.01  | 2.94  | 2.90  | 2.86  | 2.83  | 2.79  | 2.75  | 2.71     |
| 10                                   | 4.96                         | 4.10  | 3.71  | 3.48  | 3.33  | 3.22  | 3.14  | 3.07  | 3.02  | 2.98  | 2.91  | 2.85  | 2.77  | 2.74  | 2.70  | 2.66  | 2.62  | 2.58  | 2.54     |
| 11                                   | 4.84                         | 3.98  | 3.59  | 3.36  | 3.20  | 3.09  | 3.01  | 2.95  | 2.90  | 2.85  | 2.79  | 2.72  | 2.65  | 2.61  | 2.57  | 2.53  | 2.49  | 2.45  | 2.40     |
| 12                                   | 4.75                         | 3.89  | 3.49  | 3.26  | 3.11  | 3.00  | 2.91  | 2.85  | 2.80  | 2.75  | 2.69  | 2.62  | 2.54  | 2.51  | 2.47  | 2.43  | 2.38  | 2.34  | 2.30     |
| 13                                   | 4.67                         | 3.81  | 3.41  | 3.18  | 3.03  | 2.92  | 2.83  | 2.77  | 2.71  | 2.67  | 2.60  | 2.53  | 2.46  | 2.42  | 2.38  | 2.34  | 2.30  | 2.25  | 2.21     |
| 14                                   | 4.60                         | 3.74  | 3.34  | 3.11  | 2.96  | 2.85  | 2.76  | 2.70  | 2.65  | 2.60  | 2.53  | 2.46  | 2.39  | 2.35  | 2.31  | 2.27  | 2.22  | 2.18  | 2.13     |
| 15                                   | 4.54                         | 3.68  | 3.29  | 3.06  | 2.90  | 2.79  | 2.71  | 2.64  | 2.59  | 2.54  | 2.48  | 2.40  | 2.33  | 2.29  | 2.25  | 2.20  | 2.16  | 2.11  | 2.07     |
| 16                                   | 4.49                         | 3.63  | 3.24  | 3.01  | 2.85  | 2.74  | 2.66  | 2.59  | 2.54  | 2.49  | 2.42  | 2.35  | 2.28  | 2.24  | 2.19  | 2.15  | 2.11  | 2.06  | 2.01     |
| 17                                   | 4.45                         | 3.59  | 3.20  | 2.96  | 2.81  | 2.70  | 2.61  | 2.55  | 2.49  | 2.45  | 2.38  | 2.31  | 2.23  | 2.19  | 2.15  | 2.10  | 2.06  | 2.01  | 1.96     |
| 18                                   | 4.41                         | 3.55  | 3.16  | 2.93  | 2.77  | 2.66  | 2.58  | 2.51  | 2.46  | 2.41  | 2.34  | 2.27  | 2.19  | 2.15  | 2.11  | 2.06  | 2.02  | 1.97  | 1.92     |
| 19                                   | 4.38                         | 3.52  | 3.13  | 2.90  | 2.74  | 2.63  | 2.54  | 2.48  | 2.42  | 2.38  | 2.31  | 2.23  | 2.16  | 2.11  | 2.07  | 2.03  | 1.98  | 1.93  | 1.88     |
| 20                                   | 4.35                         | 3.49  | 3.10  | 2.87  | 2.71  | 2.60  | 2.51  | 2.45  | 2.39  | 2.35  | 2.28  | 2.20  | 2.12  | 2.08  | 2.04  | 1.99  | 1.95  | 1.90  | 1.84     |
| 21                                   | 4.32                         | 3.47  | 3.07  | 2.84  | 2.68  | 2.57  | 2.49  | 2.42  | 2.37  | 2.32  | 2.25  | 2.18  | 2.10  | 2.05  | 2.01  | 1.96  | 1.92  | 1.87  | 1.81     |
| 22                                   | 4.30                         | 3.44  | 3.05  | 2.82  | 2.66  | 2.55  | 2.46  | 2.40  | 2.34  | 2.30  | 2.23  | 2.15  | 2.07  | 2.03  | 1.98  | 1.94  | 1.89  | 1.84  | 1.78     |
| 23                                   | 4.28                         | 3.42  | 3.03  | 2.80  | 2.64  | 2.53  | 2.44  | 2.37  | 2.32  | 2.27  | 2.20  | 2.13  | 2.05  | 2.01  | 1.96  | 1.91  | 1.86  | 1.81  | 1.76     |
| 24                                   | 4.26                         | 3.40  | 3.01  | 2.78  | 2.62  | 2.51  | 2.42  | 2.36  | 2.30  | 2.25  | 2.18  | 2.11  | 2.03  | 1.98  | 1.94  | 1.89  | 1.84  | 1.79  | 1.73     |
| 25                                   | 4.24                         | 3.39  | 2.99  | 2.76  | 2.60  | 2.49  | 2.40  | 2.34  | 2.28  | 2.24  | 2.16  | 2.09  | 2.01  | 1.96  | 1.92  | 1.87  | 1.82  | 1.77  | 1.71     |
| 26                                   | 4.23                         | 3.37  | 2.98  | 2.74  | 2.59  | 2.47  | 2.39  | 2.32  | 2.27  | 2.22  | 2.15  | 2.07  | 1.99  | 1.95  | 1.90  | 1.85  | 1.80  | 1.75  | 1.69     |
| 27                                   | 4.21                         | 3.35  | 2.96  | 2.73  | 2.57  | 2.46  | 2.37  | 2.31  | 2.25  | 2.20  | 2.13  | 2.06  | 1.97  | 1.93  | 1.88  | 1.84  | 1.79  | 1.73  | 1.67     |
| 28                                   | 4.20                         | 3.34  | 2.95  | 2.71  | 2.56  | 2.45  | 2.36  | 2.29  | 2.24  | 2.19  | 2.12  | 2.04  | 1.96  | 1.91  | 1.87  | 1.82  | 1.77  | 1.71  | 1.65     |
| 29                                   | 4.18                         | 3.33  | 2.93  | 2.70  | 2.55  | 2.43  | 2.35  | 2.28  | 2.22  | 2.18  | 2.10  | 2.03  | 1.94  | 1.90  | 1.85  | 1.81  | 1.75  | 1.70  | 1.64     |
| 30                                   | 4.17                         | 3.32  | 2.92  | 2.69  | 2.53  | 2.42  | 2.33  | 2.27  | 2.21  | 2.16  | 2.09  | 2.01  | 1.93  | 1.89  | 1.84  | 1.79  | 1.74  | 1.68  | 1.62     |
| 40                                   | 4.08                         | 3.23  | 2.84  | 2.61  | 2.45  | 2.34  | 2.25  | 2.18  | 2.12  | 2.08  | 2.00  | 1.92  | 1.84  | 1.79  | 1.74  | 1.69  | 1.64  | 1.58  | 1.51     |
| 60                                   | 4.00                         | 3.15  | 2.76  | 2.53  | 2.37  | 2.25  | 2.17  | 2.10  | 2.04  | 1.99  | 1.92  | 1.84  | 1.75  | 1.70  | 1.65  | 1.59  | 1.53  | 1.47  | 1.39     |
| 120                                  | 3.92                         | 3.07  | 2.68  | 2.45  | 2.29  | 2.17  | 2.09  | 2.02  | 1.96  | 1.91  | 1.83  | 1.75  | 1.66  | 1.61  | 1.55  | 1.50  | 1.43  | 1.35  | 1.25     |
| $\infty$                             | 3.84                         | 3.00  | 2.60  | 2.37  | 2.21  | 2.10  | 2.01  | 1.94  | 1.88  | 1.83  | 1.75  | 1.67  | 1.57  | 1.52  | 1.46  | 1.39  | 1.32  | 1.22  | 1.00     |

## Part I Solutions

1. C  $P(L \cup D) = P(L) + P(D) - P(L \cap D) = 0.37 + 0.11 - 0.03 = 0.45$ . Here L is for Los Angeles and D is for 30-60 days overdue
2. C  $P(D_1 | B) = \frac{P(D_1 \cap B)}{P(B)} = \frac{0.15}{0.27} = 0.556$
3. D
4. B A categorical variable with K levels needs K-1 dummy variables.
5. D
6. D Two-sided test. p-value =  $2 * P(z < -1.87) = 0.06$ .
7. A  $P(\bar{X} \leq 50) = P\left(Z \leq \frac{50 - 42}{30/\sqrt{36}}\right) = P(Z \leq 1.6) = 0.9452$ .
8. D  $\left(\frac{z_{\alpha/2}}{E}\right)^2 p(1-p) = \left(\frac{2.326}{0.02}\right)^2 0.45(1-0.45) = 3348.61 \approx 3349$ . If use 2.33 as critical value, the solution is 3360. Choose the closest one from these 4 options.
9. D  $np_0 = 75 \geq 5$ ,  $n(1-p_0) = 25 \geq 5$ . Therefore,  

$$z = \frac{\bar{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.68 - 0.75}{\sqrt{\frac{0.75(1-0.75)}{100}}} = -1.62$$
Do not reject null hypothesis  
Here  $H_0: p = 0.75$ ;  $H_a: p \neq 0.75$ , where  $p$  is the population proportion of restaurant employees affected by work stress.
10. A The expected monetary values are shown in the table below. Choose Alternative 1

|               | Light Demand | Medium Demand | High Demand | EMV  |
|---------------|--------------|---------------|-------------|------|
| Alternative 1 | 18           | 28            | 20          | 21.6 |
| Alternative 2 | 25           | 17            | -5          | 13.6 |
| Alternative 3 | 3            | 40            | 16          | 18   |
| Alternative 4 | 5            | 40            | 20          | 20   |
| Probability   | 0.4          | 0.3           | 0.3         |      |
11. A Expected value with Perfect Information =  $25 \times 0.4 + 40 \times 0.3 + 20 \times 0.3 = 28$ . Therefore, EVPI =  $28 - 21.6 = 6.4$
12. C See contingency table below

|          | Low Volume                    | High Volume                   | Total   |
|----------|-------------------------------|-------------------------------|---------|
| Dies     | $0.031 \times 0.23 = 0.00713$ | $0.007 \times 0.77 = 0.00539$ | 0.01252 |
| Survives | $0.23 - 0.00713 = 0.22287$    | $0.7 - 0.00539 = 0.69461$     | 0.91748 |
| Total    | 0.23                          | 0.77                          | 1.000   |



13. B  $= \text{BINOMDIST}(2, 100, 0.01252, 0) = \frac{100 \times 99}{2 \times 1} (0.01252)^2 (0.98748)^{98} \approx 0.2257$ .
14. B  $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{71.2 - 69}{8/\sqrt{41}} = 1.761$ . Critical value with DF=40 at  $\alpha = 0.01$  is 2.4; Critical value with DF=40 at  $\alpha = 0.05$  is 1.684; Critical value with DF=40 at  $\alpha = 0.1$  is 1.303. Reject  $H_0$  at  $\alpha = 0.05$  or  $\alpha = 0.1$ , can not reject  $H_0$  at  $\alpha = 0.01$ .
15. B  $\mu_0 + t_{0.05, 40} \frac{s}{\sqrt{n}} = 69 + 1.684 * \frac{8}{\sqrt{41}} = 71.104$ .

## Part II Solutions

1.
  - a.  $H_0 : p \leq 0.20$ ,  $H_a : p > 0.20$ . Here  $p$  refers to the proportion of customers interested in self-checkout lines. (2 pts. It's OK if  $H_0: p = 0.2$ .)
  - b.  $z = \frac{\bar{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$ . Follows approximately standard normal distribution if  $p = p_0$  and  $np_0 = 3469.2 \geq 5, n(1-p_0) = 13876.8 \geq 5$ . (2 pts)
 
$$\bar{p} = \frac{3608}{17346} = 0.208$$

$$z = \frac{0.208 - 0.2}{\sqrt{\frac{0.2 * 0.8}{17346}}} = \frac{0.008}{0.0031} = 2.58$$
. (3 pts. Give 1 pt for correct  $\bar{p}$ , 1 pt for correct formula of  $z$ , 1 pt for correct standard error.)
  - c. p-value =  $P(Z > 2.58) = 1 - 0.9951 = 0.0049$ . (2 pts)
 

Since  $0.0049 < 0.05$ , we reject  $H_0$ . This indicates that the sample provides evidence that consumer interest is more than 20%. Therefore, the CEO should operate the self-checkout lines. (1 pt)
2.
 

$H_0: \mu_1 - \mu_2 \leq 0$ ;  $H_a: \mu_1 - \mu_2 > 0$ . Here  $\mu_1, \mu_2$  refers to the population mean of the copies printed by the name-brand and generic cartridge, respectively. (2 pts)

This is a paired t-test. Test statistic

$$t = \frac{19.34 - 0}{\frac{63.9}{\sqrt{35}}} = \frac{19.34}{10.8} = 1.79$$
. (3 pts)
 

Since this is a one-tailed test at the  $\alpha = 0.05$  level, we need the critical value for  $df = N - 1 = 35 - 1 = 34$ . Using a t-distribution table, the critical value for a one-tailed test with 34 degrees of freedom at  $\alpha = 0.05$  is approximately 1.691. (2 pts)

Since the test statistic 1.791 is greater than the critical value 1.691, we reject the null hypothesis. There is sufficient evidence to conclude that, on average, the name-brand cartridge produces more copies than the generic cartridge. (2 pts)

3.

- a. Predicted emissions =  $9.16 - 0.226 * 20 + 0.229 * 8.2 + 1.20 * 2.066 - 0.085 * 7 - 0.00052 * 265 - 0.035 * 6 + 0.414 * 3.5 = 9.5032$ . (2 pts)  
The residual =  $9.7 - 9.5032 = 0.1968$ . (1 pt) It is OK if the solution is -0.1968.

- b.  $R^2 = \frac{SSR}{SST} = \frac{138.443}{149.216} = 0.9278$ . (2 pts)  
0.9278 indicates that approximately 92.78% of the variance in the dependent variable can be explained by the independent variable(s) in the regression model. (1 pt)

If students answer something like the following, “an R-squared value close to 1 (like 0.9278) generally indicates that the model fits the data well. It suggests that the regression model provides a strong explanation of the data, making it a potentially valuable tool for prediction and analysis”. It is also OK to give 1 pt.

- c. Denote  $\beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6, \beta_7$  as the population coefficient for the predictors *MPG, sec, liters, wt, hp, cyl, passengers*, respectively.

$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = \beta_7 = 0$ ;

$H_a$ : at least one  $\beta_i \neq 0$ . (2pts)

F test statistic =  $MSR / MSE = 19.778 / 0.414 = 47.77$ . (2pts)

The critical value at  $\alpha=0.05$  is  $F_{0.05,7,26} = 2.39$ . (1 pt)

Since the test statistic  $47.77 >$  critical value, we reject  $H_0$ . There is enough evidence to say that at least one of the predictors *MPG, sec, liters, wt, hp, cyl, passengers* is significant. (1 pt)

- d.  $0.4142 \pm t_{0.025,26} * s.e = 0.4142 \pm 2.056 * 0.2934 = (-0.1890, 1.0174)$ . (3 pts, partial credits: give 1 pt for correct t-value, 1 pt for correct mean, 1 pt for correct s.e.)  
Since 0 is within this confidence interval, we cannot reject the null hypothesis that this coefficient is 0. (1 pt)

- e.  $VIF(\text{liters}) = \frac{1}{1 - R_{\text{liters}}^2} = \frac{1}{1 - 0.902} = 10.204$  (3 pts. If students use a wrong  $R^2$  but specify the correct VIF formula, give 1.5 pts as partial credits)

- f. This is because “liters” is correlated with other predictors in the model.  $VIF(\text{liters}) > 5$ , which demonstrates multicollinearity in the dataset. (2 pts. Give full credits if they mention “multicollinearity” or “highly correlated”).